

HEAVY SUPPLIERS WAREHOUSE PROJECT KPI FRAMEWORK

15 Business KPIs — Definitions, Priority, Measurement & Project Relevance

Prepared for	Data Analysis / Power BI Project — HeavySuppliersWarehouseDatasets (CadetX Virtual Work Experience brief)
Purpose	Translate the project objectives into measurable, field-level business performance indicators built on the cleaned and validated dataset.
Scope	Inventory • Product Movement • Suppliers • Procurement • Warehouse Efficiency • Sales • Finance

1. Executive Summary

The Heavy Suppliers project is fundamentally about using data to improve warehouse and supply-chain decisions. The KPI framework therefore prioritises measures that answer five practical questions: Is inventory moving properly? Are suppliers and procurement reliable? Can customer demand be fulfilled? Are warehouses operating efficiently? Is the operation financially healthy?

The 15 KPIs below are arranged by project-business priority, established after cleaning, integration, feature engineering, and validation of the underlying data. The ranking does not mean lower-ranked KPIs are unimportant — it means the first KPIs are more directly connected to the core warehouse, inventory, supplier and procurement problems described in the project brief, and are built on fields that passed validation cleanly.

Three specific data-quality findings from the validation stage shape several KPI definitions below and are called out at the relevant KPI rather than hidden inside a formula: (1) 417 payments (2.2%) cannot be matched to one specific invoice because their invoice_id was duplicated in the source data; (2) inventory_master.current_stock is internally consistent with the stock ledger but on a different numeric scale than the table's own reorder_level/max_stock fields; (3) branches.avg_monthly_revenue does not match revenue computed from actual transactions, by a wide and inconsistent margin.

Priority	KPI	Primary question answered
1	Inventory Turnover	Is inventory moving efficiently?
2	Supplier On-Time Delivery Rate	Are suppliers delivering when promised?
3	Purchase Order Fulfillment Rate	Are procurement orders being completed?
4	Order Fulfillment Rate	Can the operation fulfil customer demand?
5	Stock Out Frequency	How often does inventory become unavailable?
6	Average Procurement Lead Time	How long does replenishment take?
7	Total Sales Revenue	How much sales value is being generated?

Priorit y	KPI	Primary question answered
8	Gross Margin %	How profitable are the sales/products?
9	Average Daily Stock Outflow	How quickly is stock leaving inventory?
10	Revenue per Employee	How efficient is workforce utilisation?
11	Order Cancellation Rate	How often do customer orders fail through cancellation?
12	Stock Value per Sqft	How much inventory value is held per unit of warehouse space?
13	Stock Units per Sqft	How much physical stock is held per unit of warehouse space?
14	Unresolved Payments	How much payment exposure remains unresolved?
15	Overdue Invoice %	How much receivables exposure is overdue?

2. KPI Definitions & Business Rationale

1. Inventory Turnover

Business area	Inventory & Product Movement
Definition	How quickly inventory is sold or used, relative to how much is held on average.
Why it matters	Central to the project because the business wants better inventory health, less excess stock, and better replenishment decisions.

2. Supplier On-Time Delivery Rate

Business area	Supplier Performance
Definition	The percentage of received POs delivered on or before the expected delivery date.
Why it matters	Shows supplier dependability and is the leading indicator for future stockouts and fulfilment problems — a late supplier shows up here before it shows up anywhere else.

3. Purchase Order Fulfillment Rate

Business area	Procurement
Definition	The percentage of purchase orders that were successfully received.
Why it matters	Shows how effectively procurement converts purchasing decisions into actual stock received.

4. Order Fulfillment Rate

Business area	Sales & Operations
Definition	The percentage of customer orders that were successfully delivered.
Why it matters	Connects inventory and warehouse capability directly to customer service and operational reliability — the core promise of the business.

5. Stock Out Frequency

Business area	Inventory Health
Definition	How often inventory reaches an agreed stockout condition.
Why it matters	Stockouts are one of the key risks in inventory management; this KPI assesses whether replenishment is preventing availability problems.

6. Average Procurement Lead Time

Business area	Procurement
Definition	Average number of days from placing a PO to receiving it.
Why it matters	Determines when to reorder and how much buffer procurement needs to replenish stock without a stockout.

7. Total Sales Revenue

Business area	Sales & Demand
Definition	Total monetary value of qualifying sales during the reporting period.
Why it matters	Provides the commercial context for product movement and allows performance to be analysed by branch, product, and channel.

8. Gross Margin %

Business area	Profitability
Definition	Percentage of sales remaining after product cost is deducted.
Why it matters	Prevents the project from focusing only on sales volume; identifies which products generate revenue versus which contribute real gross profit.

9. Average Daily Stock Outflow

Business area	Product Movement
Definition	Average quantity of stock leaving inventory per day through valid OUT movements.
Why it matters	Shows the pace of product movement and supports demand estimation and replenishment planning.

10. Revenue per Employee

Business area	Warehouse Efficiency
Definition	Revenue generated relative to the number of employees.
Why it matters	Supports branch and warehouse efficiency comparisons and helps management understand revenue relative to workforce size.

11. Order Cancellation Rate

Business area	Sales & Operations
Definition	Percentage of customer orders that were cancelled.
Why it matters	A high rate can point to inventory availability, operational, customer, or process issues, and should be investigated by branch, channel, and product.

12. Stock Value per Sqft

Business area	Warehouse Space Utilisation
Definition	Inventory monetary value held per square foot of warehouse area.
Why it matters	Measures how much inventory value is being held in available warehouse space and directly supports space-utilisation analysis — a project goal with no other supporting KPI.

13. Stock Units per Sqft

Business area	Warehouse Space Utilisation
Definition	Number of inventory units held per square foot of warehouse area.
Why it matters	Complements Stock Value per Sqft by showing physical stock density rather than dollar density — a branch can be dense in value but sparse in units, or the reverse.

14. Unresolved Payments

Business area	Financial Operations
Definition	Number or value of invoices that are not fully settled.
Why it matters	Shows outstanding payment exposure and adds financial visibility to the operational dashboard.

15. Overdue Invoice %

Business area	Financial Operations
Definition	Percentage of applicable invoices that are overdue and not fully paid as of a defined reporting date.
Why it matters	Highlights receivables risk and delayed collections — the most direct cash-flow warning signal in the dataset.

End of KPI Framework